

Putting the Count Back Into Accountability

An Analysis of Transparency Data About the Sexual Exploitation of Minors

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Alarmist and sensationalist statements about the “explosion” of online child sexual exploitation or CSE dominate much of the public discourse about the topic. Based on a new dataset collecting the transparency disclosures for 16 US-based internet platforms and the national clearinghouse collecting legally mandated reports about CSE, this study seeks answers to two research questions: First, what does the data tell us about the growth of online CSE? Second, how reliable and trustworthy is that data? To answer the two questions, this study proceeds in three parts. First, we leverage a critical literature review to synthesize a granular model for CSE reporting. Second, we analyze the growth in CSE reports over the last 25 years and correlate it with the growth of social media user accounts. Third, we use two comparative audits to assess the quality of transparency data. Critical findings include: First, US law increasingly threatens the very population it claims to protect, i.e., children and adolescents. Second, the rapid growth of CSE report over the last decade is linear and largely driven by an equivalent growth in social media user accounts. Third, the Covid-19 pandemic had no statistically relevant impact on report volume. Fourth, while half of surveyed organizations release meaningful and reasonably accurate transparency data, the other half either fail to make disclosures or release data with severe quality issues.

Additional Key Words and Phrases: social media, transparency reporting, audit, child sexual abuse materials, National Center for Missing and Exploited Children, CyberTipline

1 INTRODUCTION

Public debate about child sexual exploitation (CSE) features many adamant to alarming claims. There are the headlines in national newspapers announcing that “the internet is overrun with images of child sexual abuse” [66], “Instagram connects vast pedophile network” [57], and “AI is about to make the online child sex abuse problem much worse” [106]. There are Senate hearings on “big tech and the online child sexual exploitation crisis” featuring a who-is-who of social media CEOs [24] and laws referencing the “explosion in the multijurisdictional distribution of child pornography” [134]. There also are the Moms for Liberty, Qanon adherents, and Proud Boys mobilizing to “#SaveTheChildren” while maligning everyone else, particularly LGBT folk, as “pedos” or “groomers” [11, 82]. By contrast, experts such as academics [116], lawyers [50], hotline operators [104], and police [29, 63] tend to blame the Covid-19 pandemic, often in unusually lurid terms.

Given the gravity of the topic and the creeping sensationalism infusing even the law, two questions stand out:

- (1) What does available data tell us about the extent of online child sexual exploitation?
- (2) How accurate and trustworthy is that data?

This article presents the results of a study that seeks to answer these two research questions based on the transparency disclosures (or lack thereof) of 15 technology firms, including popular social media platforms, and 2 not-for-profits, including the national clearinghouse for reports about CSE, the National Center for Missing and Exploited Children or NCMEC. According to US law, service providers needn’t proactively scan for child sexual abuse materials (CSAM) and other exploitative activities. But when they become aware of them, they must file so-called CyberTipline reports in an expedient manner.

The study comprises three parts, one to establish context and one each to answer the two research questions. In particular, the first part is based on a critical literature review that draws on a broad spectrum of sources not limited

to the academic literature. The primary goals were to synthesize a conceptual model of CyberTipline reporting and to identify gaps in our knowledge. The second part is an analysis of the yearly CyberTipline report volume over the last quarter century. That includes determining a suitable regression model and identifying the likely primary force driving the rapid growth over the last decade. The third part features two comparative audits of transparency data. Both audits leverage redundant or repeated disclosures of what should be the same quantities to assess data quality. At the same time, they differ significantly in what constitutes (un)acceptable divergence.

Significant findings include:

- (1) A significant overreach of US law that increasingly threatens the very population it claims to protect, i.e., children and adolescents;
- (2) Linear growth in CyberTipline reports over the last decade that is largely driven by a corresponding increase in social media user accounts;
- (3) No statistically relevant impact of the pandemic on report volume;
- (4) A half-half split between firms that release meaningful and reasonably accurate transparency data and those that either fail to make disclosures or release data with severe quality issues.

The second finding directly contradicts claims of an “explosion” of CSAM or an internet “overrun with images of child sexual abuse.” But that doesn’t change that the deluge of reports is bringing NCMEC close to breaking point [48]. As discussed in Section 5, the findings of this study not only allow us to provide a bound on possible future growth. But they also suggest two approaches for reigning in the deluge of CyberTipline reports. Alas, evaluating their impact requires further research and more granular transparency data.

All study data as well as the Python and R code used for analysis have been released under a permissive license. Together with an extended technical report, they have been archived in an academic research repository [47].

2 METHODS

To establish necessary context, this study starts out with a critical literature review. It is based on a mix of broad and specifically targeted searches of both internet and academic repositories that iteratively refine prior results. The goal is to synthesize a realistic conceptual model of CyberTipline reporting that captures the relationship between illegal, detected, reported, and uniquely actionable incidents and offers granular detail even where existing transparency practices do not. Since more than 99% of all CyberTipline reports concern CSAM, the literature review paid particular attention to breaking down this monolithic category into subcategories reflecting kinds of CSAM observed in practice. The iterative expansion of sources considered as part of the review ensures that those subcategories are representative, though it cannot guarantee that they are exhaustive, i.e., complete. Where data is available, we incorporate it into the review. However, the review also surfaced several aspects that would benefit from quantification.

2.1 Data

The subsequent parts of the study build on a new dataset comprising quantitative CSE transparency disclosures for about thirty organizations, covering large technology firms, their subsidiaries and social media brands, as well as a couple of not-for-profits [47]. The latter include the National Center for Missing and Exploited Children or NCMEC, which serves as clearinghouse for reports about CSE, which are mandated by US law and received through its CyberTipline. Since social media thrive on user-generated content, the primary criterion for inclusion in the dataset—in

Table 1. An overview of surveyed organizations and their subsidiaries or brands. Criteria for inclusion are social media (Soc), very large platforms (VL), and platforms that filed $\geq 100k$ reports (.1M)

Organization	Soc	VL	.1M	Disclosures (Months)	Audit
Amazon	✓	✓		2020..	12 Reports
Twitch	✓			H1 2020..	6
Apple		✓			—
Automattic	✓				—
Tumblr	✓				
Aylo née MindGeek		✓	✓		Reports
Pornhub		✓	✓	2020..	6
Discord	✓		✓	H2 2020..	3 —
Google	✓	✓	✓	H1 2020..	6 Reports
YouTube	✓		✓	H1 2020..	6
Meta	✓	✓	✓		Data
Facebook	✓		✓	Q3 2018..	3
Instagram	✓		✓	Q2 2019..	3
WhatsApp	✓		✓		
Microsoft	✓	✓	✓	H1 2020..	6 Reports
LinkedIn	✓	✓	✓	H1 2019..	6
Skype	✓				
Teams	✓				
NCMEC				2019..	12 Reports
Omegle			✓		—
Pinterest	✓	✓		H1 2020..	3 Reports
Quora	✓				—
Reddit	✓		✓	2019..	6 Reports
Snap	✓	✓	✓	H2 2019..	6 Reports
TikTok	✓	✓	✓	Q1 2022..	3 Data
Wikimedia		✓			—
X	✓	✓	✓	H1 2024..	6 —
Twitter	✓		✓	H2 2018..H1 2021	

addition to participating in CyberTipline reporting—is being a popular social media site based on Buffer’s yearly rankings from 2022 through 2024 [78, 79, 105]. To ensure broad coverage, further criteria are being labelled a very large online platform by the European Commission [27] and having filed 100,000 or more CyberTipline reports during any year, which are roughly 0.3% of the total report count for 2023.

Where available, an organization’s entry in the dataset includes the quantitative disclosures, preserving their numerical precision, unit names (with some well-defined exceptions for common metrics), and reporting periods. At the same time, some organizational entries record only the absence of relevant transparency data, e.g., Apple, or the relationship to other entries that do contain data, e.g., Meta’s entry lists Facebook, Instagram, and WhatsApp as corporate brands. Table 1 lists the 15 corporations and 2 not-for-profits surveyed by this study, the criteria for inclusion, the (indented) subsidiaries/brands meeting the criteria, time periods covered by their transparency disclosures, current disclosure frequencies in months, and audits applied to their data. Amongst the 15 corporations, Omegle is probably the most obscure; the video chat service shut down in November 2023 to settle a lawsuit filed by a minor who was sexually

exploited on that platform. As shown in the table, CyberTipline report counts were audited for seven corporations plus NCMEC, and historical data disclosures were audited for another two corporations.

The dataset further includes several supplemental tables and code for more easily processing the data. Supplemental tables comprise domain-specific information, such as the original machine-readable disclosures by Discord, Meta, Microsoft, and TikTok, as well as generally useful information, such as per-country population counts from the UN Population Division [132] and global vector outlines from Natural Earth.¹ Since Meta updates the data but not URL for its quarterly csv disclosures, the data for Q2 2021 through Q1 2022 (inclusive) was accessed through the Internet Archive.² While reported statistics go as far back as Q3 2018, no csv files from before Q2 2021 are available. To help clean up, validate, and enrich the data, the code includes a 4,400-line Python library. It also includes the Python and R notebooks with the analysis for this study.

As illustrated by the *Disclosures* column in Table 1, most organizations release some transparency data. At the same time, NCMEC’s disclosures are the by far most comprehensive, thanks largely to two reports to the Congressional Committees of Appropriations for the years 2022 [99] and 2023 [101]. Mandated by the explanatory statement for the Consolidated Appropriations Act of 2022 [20], the data included in these disclosures goes beyond the per-provider and per-country breakdowns published before and provides, for the first time, comprehensive statistics on reported activities, the uniqueness of reports and attachments, as well as the relationship between perpetrators and abused minors. While most of the data covers recent years only, i.e., from 2020 forward, the disclosures do include yearly CyberTipline report counts going back to the beginning in 1998. These statistics are the foundation for the second part of the study, which considers the volume of CyberTipline reports.

2.2 Audits

To assess the quality of technology industry transparency efforts, this study includes two audits of transparency data as its third part. The first and primary audit compares CyberTipline report counts for electronic service providers or *esps* as disclosed by their producers and their only consumer. While almost all service providers that disclose counts do so semiannually or quarterly and several of them distinguish between subsidiaries or brands, NCMEC’s yearly disclosure frequency and corporation-only breakdown for 2019 and 2020 determine the granularity for all audits, i.e., yearly per-corporation counts. Out of the 16 organizations other than NCMEC contained in the dataset, seven include CyberTipline report counts in their transparency disclosures and hence are subject to the first audit. Twitter’s transparency disclosures only covered the number of unique accounts closed. However, X’s disclosures for H1 2024, for the first time, do include CyberTipline report counts, which raises the possibility of inclusion in a future iteration of this audit.

The second audit compares computer-readable data disclosures, e.g., csv files, by the same service provider for subsequent reporting periods. In particular, for each reporting period $p + 1$, it compares all entries shared with the previous reporting period p , which should be the entries for periods 1 to p or all entries in the file for reporting period p . The dataset contains two corporations that release current and past transparency data in machine-readable formats and hence are subject to the second audit. They also happen to be distinct from the seven corporations covered by the first audit.

Since most surveyed organizations engage in surveillance capitalism [139], i.e., critically depend on their ability to process and analyze large volumes of data, we might expect to mostly find pairwise identical quantities. Furthermore, in the rare case that quantities differ, i.e., exhibit some kind of error, we’d expect that either service provider

¹<https://www.naturalearthdata.com>

²https://web.archive.org/web/20240000000000*/https://transparency.fb.com/sr/community-standards/

or NCMEC would have noticed and provided a credible explanation with the data. Alas, in the real world, instances of divergent transparency data aren't hard to come by, but apparent awareness and explanations thereof are. In fact, we encountered only one instance of a service provider explaining a major discrepancy. At the same time, the audits identified significant data quality issues for three service providers. Furthermore, claims made by two of the providers are sufficiently discordant with the data to raise concerns about the veracity of their transparency reporting.

With numerical equality too blunt an audit instrument, the challenge becomes to tease apart different kinds of errors. Hence we need to define what kind of errors we can reasonably expect for each audit. For CyberTipline report counts, we are comparing the results of two independent organizations tabulating the same unidirectional message flows. It seems reasonable to allow for small errors in the classification and tabulation of these messages due to (non-deterministic) software bugs and hardware outages. Hence, differences between comparable report counts should be small relative to their overall magnitude, say less than 1%, and not exhibit other discernable patterns, i.e., they reflect random errors only.

For data files, we are comparing the same metrics published by the same organization some number of months apart. In this case, values should only change as a result of deliberate modification, notably, to correct incomplete or buggy disclosures. Hence, differences should exhibit temporal or logical locality and hence qualify as systemic errors only. When considering both audits, we note an altogether remarkable difference: While small random errors are acceptable for the first audit, they are unacceptable for the second audit. Likewise, while small systemic errors are acceptable for the second audit, they are unacceptable for the first audit.

2.3 Statistics

We did consider the intra-class correlation coefficient for count data for comparing the tallies of CyberTipline reports [74]. However, due to the small number of per-year, per-provider samples, with $2 \leq n \leq 5$ and $\bar{n} = 3.86$, we could only compute a goodness-of-fit for the entire dataset, across all providers, with $N = 27$. Those calculations demonstrate that neither the Poisson distribution nor a negative binomial distribution with variance growing linearly with mean were suitable models. A negative binomial distribution with variance growing quadratically with mean fared better but nonetheless couldn't account for all of the samples. Since the results offer little insight beyond what can already be gleaned by inspection, we decided to drop this marginal statistic.

Instead, we are basing our comparison on Bland-Altman or mean-difference plots [4] for each of the seven providers that publish CyberTipline report counts as well as for all of them together. Consistent with the above stated assumption that the error for CyberTipline report counts grows with magnitude, we plot percentage differences, not absolute values. We use the mean as denominator, since it provides an unbiased best guess for the true value in the absence of other information. Since the normality assumption does not hold for count data, we omit the customary 95% limits of agreement from the plots and instead trace the mean difference. We then rely on visual inspection to analyze patterns and outliers, while also employing the sign test and Fisher's exact test to confirm their statistical significance [65].

2.4 Global Impact and Limitations

Even though this study is based on a mechanism mandated by us law, the reach of American internet services all but ensures that the resulting data is of global significance. Notably, Meta generated more than 84% of all CyberTipline reports every year since 2019 and more than 91% from 2019 to 2021 (inclusive). Furthermore, based on Meta's "monthly active people" [33, 85, 86], i.e., users that logged into Facebook, Instagram, Messenger, or WhatsApp at least once during

a month, and UN population estimates [132], Meta’s user share amongst the global population, across *all* ages, grew from 37% in 2019 to 49% in 2023!

That reach, however, is not uniform, with China being the most significant holdout. In fact, none of the social media platforms operated by the corporations included in this study are accessible from within China. Though Microsoft’s LinkedIn did have a local equivalent, and TikTok, in addition to being under Chinese ownership, does have a local equivalent. Furthermore, while Apple’s iCloud and iMessage are available within China, they use local servers that are not operated by Apple. Amongst several others, Belarus, India, Iran, Pakistan, and Russia similarly restrict at least some American social media [42].

Despite the uneven global reach, CyberTipline reporting has significant impact on countries where American service providers have a substantial presence. For example, in its transparency disclosures, Germany’s federal police repeatedly acknowledges NCMEC’s CyberTipline as the by far most significant source for reports about CSE, though it does not quantify their share [10]. However, this study only considers NCMEC’s per country statistics. Analyzing the disclosures of international police forces is an interesting and complimentary topic for future work.

This study focuses on centralized corporate social media because they currently are the dominant means for accessing user-generated content. That is highly effective, with the surveyed 16 out of almost 250 organizations accounting for more than 97% of all CyberTipline reports every year. Since these social media are typically accessed through mobile devices, this study also includes the currently dominant means for accessing CSAM [123].

To put it differently, this study only marginally covers formerly dominant but still actively used technologies, such as peer-to-peer networks and the dark web [123], as well as possible future dominant technologies, such as federated social media. To begin with, most dark websites and some peer-to-peer networks aren’t particularly concerned with legal compliance and thus highly unlikely to file CyberTipline reports. Meanwhile, many federated social media instances may take their content moderation and legal compliance more seriously, but they nonetheless won’t (be able to) allocate resources to trust and safety that are even remotely comparable to centralized corporate social media. As a case in point, Mastodon gGmbH operates the largest Mastodon instance with 240,000 active users³ and also performs content moderation. But even though CSAM is being traded across the so-called fediverse [128], the firm filed a mere 16 CyberTipline reports in 2023. By comparison, the bottom three organizations surveyed by this study are the Wikimedia Foundation with 34 reports, Apple with 267 reports, and Aylo with 2,596 reports.

That is not to suggest that the relationship between corporate social media and the volume of CyberTipline reports can be reduced to anything as quaint as the allocation of capital. On the contrary, the findings, particularly Section 4.2, raise the distinct possibility that it took the combination of surveillance-capitalist social media with their amplification of base human emotions and potential for near-global virality, indiscriminate and overreaching us laws, and cell-phone wielding teenagers to create something akin to a perfect storm that keeps producing an ever worsening deluge of CSAM. Remarkably, that same data also suggests that CSE is not nearly as terrible a problem as NCMEC’s yearly statistics suggest.

3 BACKGROUND

3.1 US Law

Chapter 110 of 18 US Code [135], the United States’ criminal code, concerns the “sexual exploitation and other abuse of children.” It prohibits grooming children (§2251), selling or buying children (§2251A), the production, distribution, and possession of imagery depicting minors engaging in “sexually explicit conduct” (§2252) and any other materials

³<https://mastodon.fediverse.observer/list>

qualifying as “child pornography” (§2252A) even in other countries (§2260). Further prohibitions apply to tricking minors into accessing harmful materials with misleading domain names (§2252B) or page contents (§2252C). As long as providers of electronic services immediately report such activities and materials to NCMEC’s CyberTipline (§2258A), it explicitly limits their civil and criminal liability (§2258B).

Penalties include criminal (§2253) and civil forfeiture (§2254) as well as mandatory restitution (§2259) in addition to fines and long prison sentences, e.g., 5–20 years for the first violation of §2252A and 15–40 years for subsequent violations. The only violations of §2252 and §2252A without a minimum penalty are possession of CSAM. According to the guidelines of the US Sentencing Commission [110], a first-time offender possessing less than 10 images, none of them involving prepubescent children, sadomasochism, or violence, faces between 27–33 months in prison in addition to a \$10k–\$100k fine. However, thanks to a 1996 amendment, if the offense involves a computer, the penalty increases to 33–41 months in addition to a fine of \$15k–\$150k. Still assuming less than 10 images, no prepubescent children, s/m, or violence, if the offense involves offering/sending such imagery to others, the penalty increases to 51–63 months. If it also involves “pecuniary gain,” “valuable consideration,” or a minor as recipient, the penalty further increases to 87–108 months.

In addition to enshrining the reporting requirement for service providers into law, Chapter 110 also prescribes NCMEC’s role as clearinghouse. Notably, it compels the center to triage and forward CyberTipline reports to law enforcement (§2258A), authorizes it to collect image and video hashes and share that information with service providers (§2258C), and exempts the organization from any civil and criminal liability that might arise from running the CyberTipline (§2258D).

Despite having been created through a Congressional Act and receiving most of its funding from the US federal government, NCMEC nonetheless is a private not-for-profit corporation with an independent board. As such, it offers other, related services, such as case management for missing children, sex offender tracking, and family advocacy [38]. Still, fulfilling its role as clearinghouse necessitates close collaboration with law enforcement. Two federal appeals courts have found that NCMEC qualifies as a government agent. Hence, its inspection of CyberTipline report attachments during triage may amount to illegal warrantless searches, which may make evidence gathered during such inspections inadmissible in court [48].

3.2 CyberTipline Reports

Each CyberTipline report describes an individual incident. Accordingly, the schema starts with a summary that identifies the type as well as date and time of the incident [102]. It is followed by sections that capture granular information about involved internet services, interactions with law enforcement, the reporter of the incident, the reported person, the intended recipient, and the child victim. File attachments include violative materials such as pictures and videos that qualify as CSAM as well as other supplementary content.

A few fields are clearly designed to guide triage. The report summary includes an `escalateToHighPriority` flag and the `sextortion`, `csamSolicitation`, `minorToMinorInteraction`, and `spam` annotations. The metadata for attachments includes tags such as `potentialMeme`, `viral`, `possibleSelfProduction`, `liveStreaming` and `generativeAi`. It also includes the Tech Coalition classification [127]. That industry organization’s labelling scheme distinguishes between (A) “pre-pubescent” and (B) “post-pubescent” as well as between (1) “sex act” and (2) “lascivious exhibition” resulting in the four labels A1, A2, B1, and B2. A few actual CyberTipline reports (without attachments) have become public as part of court proceedings [93, 94].

4 FINDINGS

We present this study’s findings in three parts. Section 4.1 discusses the interaction between law and technology, introducing a model for CyberTipline reporting that goes beyond the simplistic narrative of sexual predation. When then shift focus to quantitative analysis. Section 4.2 addresses the growth of CyberTipline reports over the last quarter century, which is best modelled as two distinct epochs, the one before 2014 with very limited growth and the one since 2014 with linear but aggressive growth. Section 4.3 presents the results of the two comparative audits. It demonstrates that seven out of sixteen firms are acting in good faith and releasing reasonably accurate transparency data. One firm hasn’t released enough data yet. The remaining eight firms either are not releasing any data or have been making subpar data disclosures. Amongst the second eight, Discord, Meta, and TikTok stand out as particularly problematic.

4.1 Law and Technology

Finding 1. The prohibitions of §§2252–2252A against sexual imagery involving minors threaten severe punishment for activities that have little to do with sexual predation. They increasingly threaten the very group they are intended to protect, the minors themselves.

Findings included with bills that updated chapter 110 tell a familiar story. They emphasize a “compelling State [sic] interest in protecting children from those who sexually exploit them, and this interest extends to stamping out the vice of child pornography at all levels in the distribution chain” [134]. They further claim that “technological ease, lack of expense, and anonymity ... [of] the Internet has resulted in an explosion in the multijurisdictional distribution of child pornography” [133]. They then position the criminal law as an effective means “to dry up the market for this material by imposing severe criminal penalties” [134].

CSAM is indeed the by far most frequently reported incident type, accounting for more than 99.2% of all CyberTipline reports filed over the last four years. However, there is scant evidence for an “explosion” of CSAM. As we will demonstrate in Section 4.2, the primary driving force behind a decade of rapid and *linear* growth in CyberTipline reports is an equivalent growth in the number of social media users.

At the same time, there is clear evidence that the majority of CSAM incidents have little to do with sexual exploitation and tend towards the more benign. In 2021, Meta developed a new taxonomy of intent for sharing CSAM, distinguishing between “malicious” and “nonmalicious” users. While the first group intend to harm children, the second group “are people whose behavior is problematic and potentially harmful, but who we believe based on contextual clues and other behaviours likely did not intend to cause harm to children” [9]. When applying that taxonomy to a sample of 150 accounts that were reported to NCMEC, Meta found that malicious users are clearly distinguishable from nonmalicious ones and that “more than 75% of these did not exhibit malicious intent,” including expressions of abhorrence, inappropriate memes, and adolescents engaging in “consensual, developmentally appropriate sexual behavior” by consensually sharing sexual imagery amongst each other. The firm then leveraged these findings to roll out targeted interventions [19]. A study based on semi-structured interviews with trust and safety personnel, law enforcement officers, and NCMEC employees has since found that particularly memes are leading to “considerable inefficiencies” [48].

While the prevalence of memes is not currently quantifiable, adolescents consensually sharing sexual imagery, or “sexting,” is quantifiable and ubiquitous. Recent metastudies have put the fraction of adolescents sending or receiving sexts at, in publication order from 2018 to 2020 and 2022, 14.8% and 27.4% [81], 14% and 31% [89], as well as 19.3% and 34.8% [91]. Given these rates, it is no surprise that the lead author of the first study characterized sexting as a “normative component of teen sexual behavior and development” [71]. Technically, teenagers who do sext are violating federal

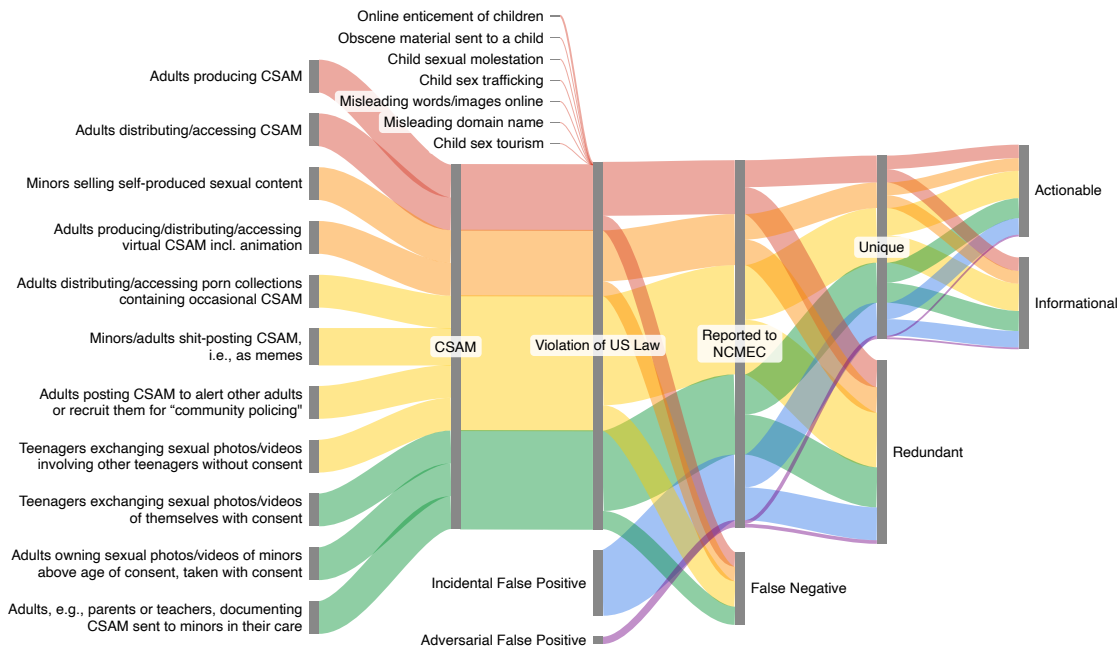


Fig. 1. The relationship between illegal, detected, reported, and uniquely actionable incidents involving the sexual exploitation of minors. The color-coding of CSAM subcategories from red to green represents a coarse classification from directly harmful to mostly harmless.

and state laws against child pornography. While an academic survey did attest significant restraint in prosecuting adolescents [137], examples for (threatened) prosecutions are not hard to come by [3, 17, 23, 37, 64, 87, 90, 131]. Some states have reduced the penalty for adolescent sexting to misdemeanors [55], but “once they have the option of lesser penalties, prosecutors are more likely to press charges” [51].

While adolescent sexting can be a perfectly healthy exploration of sexuality, it is not without significant risks. Notably, the slope from consensual to nonconsensual sharing seems slippery, with the above metastudies reporting that the prevalence for having forwarded a sext without consent is 12%, 7%, or 14.5%, respectively. Additionally, having sexted correlates with being subjugated to “any image-based sexual abuse,” increasing the prevalence from 2.8% to 37.2% amongst surveyed adolescents, a 13.2× difference [39]. Minors engage in these behaviors even though the vast majority (> 81%) know that doing so is illegal; worries about legal consequences are especially acute for boys and Black minors [129].

Figure 1 illustrates the relationship between illegal, detected, reported, and uniquely actionable incidents in Cyber-Tipline reporting. The CSAM subcategories on the left incorporate the types of incidents we just discussed and several others encountered during the literature review. We believe that the list is representative but make no claim of it being exhaustive. CSAM subcategories are color-coded from red to green according to a coarse classification of harm ranging from directly harmful down to mostly harmless to minors. Flows for subcategories are sized arbitrarily because

no reliable statistics are available. Meanwhile, category-level flows into “Violation of US Law” *are* sized according to observed CyberTipline report ratios; almost all of them are CSAM.

Finding 2. Most of the technologies used for detecting CSAM are vulnerable to adversarial attack. NCMEC is still using MD5—16 years after CERT advised to stop its use “in any capacity.”

Surveyed organizations rely on perceptual as well as cryptographic hashes for detecting known CSAM and machine learning classifiers for detecting previously unknown CSAM [76]. The critical difference between perceptual and cryptographic hashes is that the latter produce completely different values for inputs that differ by one bit only, whereas the former produce similar values for similar inputs. That makes perceptual hashes somewhat robust to changes in, say, image resolution or color grading [35]. NCMEC uses both kinds, including the MD5 and SHA-1 cryptographic hashes and Microsoft’s PhotoDNA perceptual hash [34]. With exception of Apple, Quora, and possibly Omegle, all other organizations proactively scan user-uploaded content with some perceptual hashing system, typically PhotoDNA. NCMEC and the Tech Coalition, an industry group, maintain necessary hash databases, which are accessible to industry.

In contrast to perceptual hashes, machine learning systems such as Google’s Content Safety API can detect previously unknown CSAM. Since possession of CSAM is illegal, developing such machine learning models is more complicated than for other image classifiers. Instead of directly training a model on CSAM, it is necessary to combine independently and legally trained classifiers for related traits starting with sexual activity and age [75]. Microsoft and Google make their systems available for free to qualifying organizations.

In its 2023 transparency report to the Congressional Appropriations Committees, NCMEC asserts that “images that share the same MD5 hash are identical—as are images that share the same SHA1 hash” [99]. That is false. A successful collision attack against MD5’s use in certificates has been demonstrated, and similar attacks are now practical with consumer hardware [125]. While the computational cost of attacking SHA-1 is substantially higher, collisions for SHA-1 and an attack on its use in PGP have been reported as well [77, 124]. Hence, the US government’s Computer Emergency Readiness Team (CERT) wrote in the closing days of 2008 (!) that “users should avoid using the MD5 algorithm *in any capacity*” (emphasis ours) [14], and the National Institute of Standards and Technology (NIST) started transitioning away from SHA-1 for *all* applications in 2022 [15].

Perceptual hashes don’t fare any better. For many years, PhotoDNA benefitted from security by obscurity. Since the algorithm was never published in sufficient technical detail, it also wasn’t subjugated to independent scrutiny. More recently, however, PhotoDNA has been reverse-engineered and put through its paces [72]. As it turns out, PhotoDNA as well as Meta’s PDQ [30] are vulnerable to efficient second preimage attacks, which construct images that match other images’ hashes, and also detection avoidance attacks, which cloak an image so that it does not match its hash anymore [108]. On top of that and despite Microsoft’s claims otherwise, PhotoDNA is sufficiently reversible to recreate very small thumbnail images [2]. That raises the possibility that PhotoDNA hashes are nothing other than cleverly encoded CSAM, which, of course, would make their possession illegal.

Machine learning classifiers for images aren’t more reliable either. A large number of attacks have been documented in the research literature, including both black box and white box attacks as well as attacks that falsely frame victims and that cloak illicit materials [80].

Finding 3. Automated scanning for CSAM has already been weaponized on Discord, resulting in an unknown number of account terminations.

Turning the above, mostly theoretical vulnerabilities into practical attacks and developing countermeasures are important topics for future work. But if we were to speculate about real-world attacks, one credible scenario starts with the leak of a hash database for PhotoDNA. Thereafter, those hashes are used to stage a broad denial-of-service attack on the CyberTipline reporting system by uploading borderline adult pornography modified to match leaked PhotoDNA hashes. Alternatively, hashes are used in a more targeted manner: Through social engineering, victims are manipulated into uploading seemingly innocuous images that were modified to match leaked PhotoDNA hashes. This latter attack is akin to swatting, but targets victims' internet accounts. Since many people rely on the same few accounts for a large number of services, a false flag for CSAM can have devastating consequences [54].

The latter attack also provides the basic outline for a 2023 attack against Discord users, except that no image modification was necessary [103, 138]. It involved manipulating users on Discord into posting a specific image. The image seems innocuous enough, showing an adolescent with headphones sitting in front of his computer and eating popcorn. Reliably, when users posted the image, they did not receive any promised rewards but their accounts were automatically terminated without notification or explanation. Apparently, the still image belongs to a longer video clip that shows the naked brother storming into the room and simulating intercourse on camera. Since the clip qualifies as CSAM, Discord's trust and safety team added the corresponding hashes to its database. But the firm was too aggressive and included the innocuous beginning without the naked brother, thereby creating an effective means of attacking gullible Discord users.

Discussions on Reddit and YouTube about this weaponization of falsely labelled CSAM employ a remarkable argot. Instead of using the term "child pornography," most participants substitute the initialism "cheese pizza." While the term may have originated with sexual predators, it became an important part of the PizzaGate conspiracy theory, and by now, it seems firmly entrenched with the terminally online [1].

Finding 4. Reversal rates for content takedowns by Meta and account terminations by Microsoft and Pinterest, after being actioned for CSE, have been slowly increasing over the last few years and reached 3.1%, 1.1%, and 12.8% in 2023, respectively.

Meta, Microsoft, and Pinterest are the only surveyed organizations to release statistics about reversals of content takedowns (Meta) and account terminations (Microsoft and Pinterest) for CSE. With exception of one out of four years for Meta, the reversal rate has been slowly increasing for all three firms, reaching 3.1%, 1.1%, and 12.8% in 2023, respectively. Though, with Meta's reversal rate dropping to 1.4% over the first two quarters of 2024, it looks like this trend won't continue for the firm. In any case, given the significant potential harm from being falsely accused of posting CSAM, Meta's and Microsoft's reversal rates are concerning, but Pinterest's rate is outright alarming.

Figure 1 incorporates inadvertent as well as adversarial false positives in the middle stages. It also shows false negatives, though no data is available for the rate thereof. Likewise, we have no data on actual prevalence of CSAM. Even Meta, which has been reporting prevalence rates for other categories of violative content, has not published any statistics.

Finding 5. Additional factors that contribute to high CyberTipline report counts are pictures and video that are reported more than once and low-quality reports without sufficient detail to be actionable (called *informational* by NCMEC).

As one would expect for actual abuse materials, memes, and image shared in disgust, the same pictures and videos or *pieces* are reported more than once. The multiplication factor for unique imagery (as determined by a cryptographic

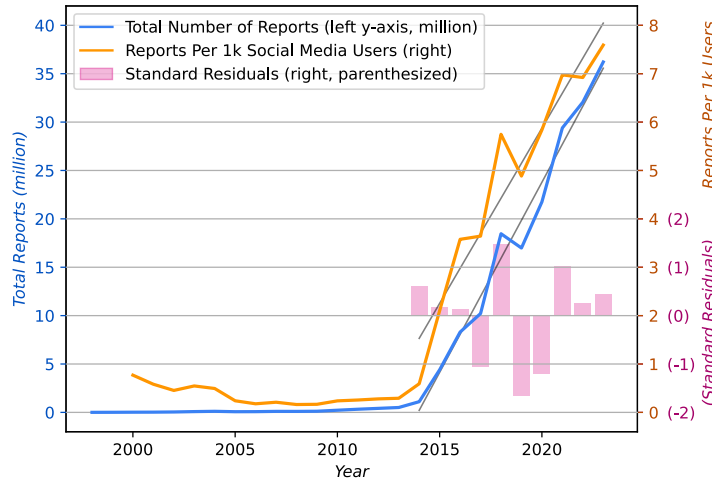


Fig. 2. Yearly CyberTipline reports (in blue, linear fit in gray, left y-axis), reports per 1,000 social media user accounts (in orange, linear fit in gray, right y-axis), and the standard residual (in magenta, right y-axis, parenthesized).

hash) and similar imagery (as determined by a perceptual hash) has been decreasing over the last four years. In 2020, identical pieces were reported $2.39\times$ on average and similar pieces $4.62\times$ on average. By 2023, those factors had come down to $2.09\times$ for identical pieces and $3.10\times$ for similar pieces. The underlying reasons for this rather substantial decrease in the degree of sharing over only four years are unclear. They might include a fracturing of the online social fabric that (say, due to polarization) reduces the reach of content in general, improvements in detection technology that reduce the reach of CSAM specifically, or increasing prevalence of content that legally counts as CSAM but isn't widely distributed, e.g., sexts. Alas, determining the root causes will require platform access to evaluate, for example, metadata.

In its transparency reports to the Congressional Appropriations Committees, NCMC remarks that the quality of CyberTipline reports differs significantly. Hence, only about half of CyberTipline reports are actionable, as opposed to what NCMC euphemistically calls “informational.” Figure 1 captures the distinction between unique and redundant attachments as well as actionable and informational reports as the final two stages on the right.

4.2 The Deluge of CyberTipline Reports

Figure 2 shows three different views onto yearly CyberTipline report counts, each with a different scale, as well as linear models for the first two views. First, the lower, blue line tracks the absolute number of reports in millions, with the left y-axis covering the range from 0 to 40 million reports. Second, the upper, orange line tracks the number of reports per 1,000 social media user accounts, with the non-parenthesized right y-axis covering the range from 0 to 8 reports per 1k users. For both curves, gray lines indicate the linear least squares fit for the decade since 2014 (inclusive). Third, the magenta bars track the standard residuals for report counts during that same decade, with the parenthesized right y-axis covering the range from -2 to 2 standard deviations. Each of the three views illustrates one of the following three findings.

Finding 6. Yearly changes in the number of CyberTipline reports are best modelled as linear within two distinct epochs, with the first ending with 2013, the second starting in 2014, and growth accelerating significantly.

A 2019 study by authors affiliated with Google, NCMEC, and Thorn characterized the growth of CyberTipline reports as exponential [12], and NCMEC has been doing the same in its 2023 and 2024 transparency reports to Congress [99, 101]. Alas, even in 2019, an exponential regression wasn't a good fit for the data. With five more years of data, it produces an even worse fit. Compared to the linear version, the exponential fit has a 15× larger mean squared error and the weighted exponential fit has a 6.5× larger error.

Finding 7. The likely primary driving force behind an absolute yearly increase of 3.93 million reports over the last decade is an equivalent growth in social media user accounts. That does not, however, account for a relative increase from roughly one to almost eight reports per 1,000 social media user accounts over the same period.

In retrospect, it is hardly surprising that social media user accounts and CyberTipline reports are closely correlated. After all, social media also file by far the most CyberTipline reports. This study, by design, includes all major US-based social media. Amongst them, Meta alone filed more than 92% of all reports from 2019 to 2021 and 85% thereafter. All social media included in this study filed between 92% and 96% of reports from 2019 to 2021. Given that, it seems safe to assume that the relationship is causal: The more users, the more users engaging in violative behavior or posting violative content.

That contention is supported by the fact that related indicators do not exhibit the same growth pattern as social media users. Notably, the number of internet users has been growing either mostly linearly or vaguely exponentially over the last 20 years, depending on source [62, 112]. Interestingly, while the ITU's global count of internet users shows mostly linear growth over the last two decades, its count of US internet users from 2005 onwards, while far noisier than social media users or CyberTipline report counts, shows the same tell-tale acceleration around 2014. That suggests that, at least for the US, social media were the prime attraction for new internet users over the last decade.

An analysis of NCMEC's by-country disaggregations [95–98, 100] offers additional but qualified support. To begin with, the analysis surfaced a few, minor data quality issues. Notably, we found entries for the same country under two labels, "French Guiana" and "Guiana, French," for the Netherlands Antilles in addition to the three island states resulting from the Antilles' breakup in 2010, for Bouvet Island even though this subantarctic Norwegian territory is an uninhabited nature preserve, and for "Europe" in addition to pretty much all European countries. With all but one count in the single digits, we simply discounted these entries for the per-capita analysis.

When ranking countries by reports per capita, we discovered an initially puzzling phenomenon: From 2019 to 2023, not only was either Libya or the United Arab Emirates (UAE) the country with the most reports per capita, but member countries of the Arab League accounted for 13, 13, 13, 9, and 14 of the top-twenty spots of countries with the most reports per capita. The substantial differences between the two top-placed countries—with Libya spending much of the 2010s in a state of civil war (ironically made worse by the UAE's interference) and the UAE the far more stable and denser populated country with better infrastructure and larger GDP—only added to the puzzlement.

DataReportal's statistics on global social media penetration point towards a simple explanation. Their ranking of countries with the most social media user accounts per capita for 2023 shows seven Arab League countries including Libya and the UAE amongst top-ten countries [69]. That neatly explains the relatively high per-capita report count for that year. However, with five and three Arab League countries amongst the top-ten for 2022 and 2021 [67, 68], the numbers do not fully align with the reports per capita. Unfortunately, DataReportal publishes only top-ten and

bottom-ten countries. Hence we do not have enough information to determine whether this is a data quality issue or reflects the impact of another, unidentified variable.

Social media user counts may be the driving force behind the increase in CyberTipline reports, but it is not the only one. As illustrated by the orange curve in Figure 2, normalized report counts have grown from less than 1 report per 1,000 users to almost 8 reports per 1,000 users over the last decade. That trend is certainly concerning, and identifying its cause is critical future work.

By comparison, incident rates for offline CSE in the US reportedly are around 1–2 incidents per 1,000 *children* (not capita) [111]. With children making up 22% of the American population [36], that corresponds to 1–2 incidents per 4,550 capita or 0.22–0.44 incidents per 1,000 capita. When using the 29% fraction of the global population as a scaling factor instead [132], the offline rate would be 1–2 incidents per 3,450 capita or 0.29–0.58 incidents per 1,000 capita.

Since the two rates are normalized by different quantities, they are not directly comparable. First, due to US privacy laws, most social media do not allow children under 13 to open accounts. Even if this minimum age is not strictly enforced and widely violated, children under 13 are going to be underrepresented amongst social media users when compared to the general population. Second, people may have accounts on more than one social media platform as well as multiple accounts per platform. For instance, one survey reports that 38% of girls 13–17 and 37% of LGBT kids have secondary or “Finsta” accounts [129]. Assuming that the undercounting and overcounting factors cancel each other out, the online prevalence of CSE appears to be an order of magnitude larger than offline activities. Unless we are inclined to believe that social media turn people into sexual predators at significant rates, that difference must be due to other factors, which is consistent with findings in the previous section.

Finding 8. The Covid-19 pandemic had no statistically relevant impact on the volume of CyberTipline reports.

As illustrated by the magenta bars in Figure 2, the standard residuals for CyberTipline reports are not particularly large, ranging from -1.7 to 1.5 standard deviations. In other words, the data is noisy when compared to its model, but there are no pronounced outliers. Furthermore, with -1.2 and +1.0, the standard residuals for the pandemic years 2020 and 2021 are clearly smaller, i.e., well within the noise. Finally, when considering year-over-year growth, the increase of +0.5 standard deviations from 2019 to 2020 actually was on the smaller end. In other words, the growth in CyberTipline report volume for the pandemic years is consistent with the growth for the rest of the decade.

We could construct more complex scenarios under which the pandemic significantly reduced organizations’ report processing capability and hence a relative increase in incidents was offset by a corresponding decrease in filed reports. But that seems unlikely because many of the claims in support of a more sustained impact do not hold up to scrutiny. Notably, besides a blog post from April 2021 by NCMC that features only year-over-year comparisons [104] and a university’s research report from May 2021 that captures the pressures faced by individual hotline staff and police officers during the pandemic [115], peer-reviewed articles [18, 50, 116] consistently cite reports by Europol from June 2020 [29] and Interpol from September 2020 [63].

Both reports mostly feature anecdotal and indirect evidence. Both reports also misrepresent data on peer-to-peer networks by the Child Rescue Coalition. They claim that graphs of sinusoidal signals that happen to rise between February and April 2020 somehow demonstrate a “significant increase in the sharing of CSEAM” (child sexual exploitation and abuse material)—to use Interpol’s boldface phrase. In addition to sharing the same sourcing and description, Interpol’s single curve very much looks like the sum of Europol’s four country-specific curves. That suggests that both police agencies are presenting the same data. However, according to y-axis labels, Interpol’s counts are roughly 5,700× larger than Europol’s aggregates.

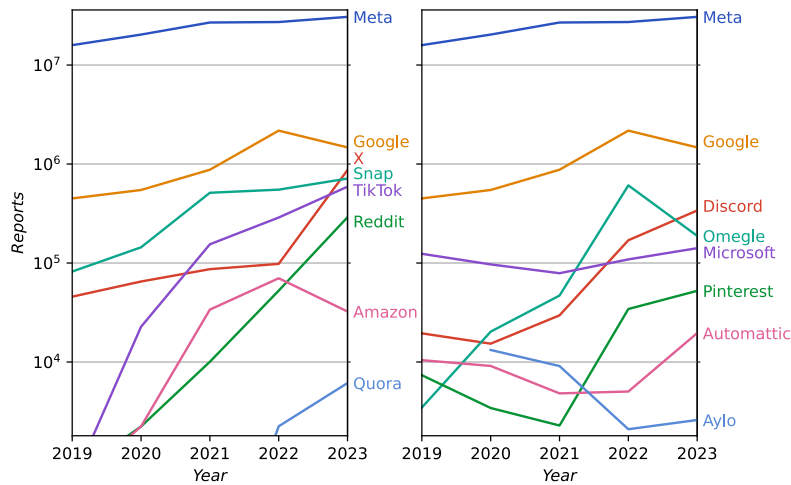


Fig. 3. Yearly CyberTipline report counts disaggregated by platform on a log scale.

At the same time, Europol’s report does contain compelling evidence that the pandemic did, in fact, result in significantly more incidents involving CSAM, albeit *at a different timescale*. When looking at CyberTipline reports NCMEC forwarded to European police during the first half of 2020 at *monthly* granularity, there is a $5\times$ spike for March 2020 only. While we concur that the pandemic is the most plausible cause, the effect is not sustained enough to have any impact on yearly statistics. Alas, since we didn’t follow a formal protocol such as PRISMA [88] in selecting publications, we cannot exclude confirmation bias for the selection of the above examples.

Before we shift focus to the two quantitative audits, we note the following negative finding:

Finding 9. Whereas the total number of CyberTipline reports per year is a reasonably good match for a linear model, that is not the case for most providers’ report counts nor for the number of attachments.

Figure 3 shows the log of yearly, per-provider report counts across two equally dimensioned grids. Report counts for Meta and Google are large and distinct enough to cleanly fit into both grids. By contrast, report counts for Apple and Wikimedia are too small to fit into the grids’ 1,800–36,000,000 reports range. All other providers appear in one of the two grids only, with placement based on a subjective judgement about readability. As the figure makes clear, per-provider report counts show significant variability and aren’t even consistently growing. Out of four providers with consistently growing report counts, Reddit has seen the most aggressive, that is, at least exponential growth.

To characterize attachment counts, we turn to the most prominent filer of CyberTipline reports, Meta. The left panel of Figure 4 shows the firm’s quarterly counts of images and videos taken down because they were classified as “child nudity & sexual exploitation” until Q1 2021 (inclusive) and as “child endangerment: sexual exploitation” thereafter. The right panel shows those same piece counts aggregated by year and juxtaposed with report counts, as disclosed by NCMEC.

At first glance, the change of metrics in Q2 2021 is curious because counts jump upwards, even though the label suggests a smaller category. Meta’s transparency report for Q2 2021 provides no explanation. However, searching for “Meta Community standards enforcement report Q2 2021” with an external search engine did produce a blog post [31] linking a PDF document with the explanation [32]. Apparently, the new metric covers not only “sexual exploitation,” but

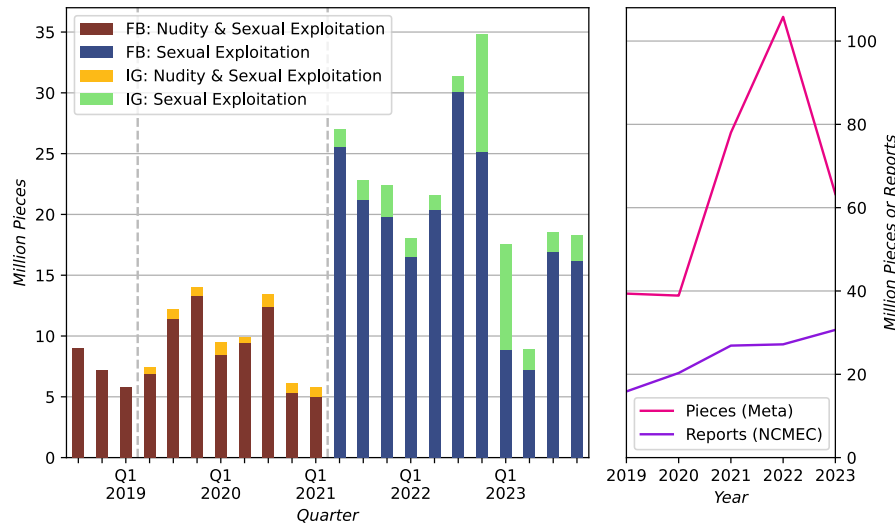


Fig. 4. Meta: The number of photos and videos, i.e., pieces, removed per quarter on the left; number of pieces and reports per year on the right.

also “sexualization of children” and “inappropriate interactions with children,” which were not tracked before. While all three are clearly related to minor safety, their aggregation make little sense. But it does explain the jump in actioned images and videos.

Even though Meta generated more than 91% of all CyberTipline reports in 2019–2021 and more than 84% in 2022–2023, the number of actioned pictures and videos does not reflect the same consistent growth year-over-year. Particularly striking are the variability between individual quarters and the substantial yearly drop in actioned pictures and videos from 2022 to 2023. One possible reason is that not all pictures and videos actioned in 2022 were severe enough to warrant reporting to NCMEC. However, even if that’s the case, the large variability is surprising. While Meta announced its intent to share more statistics about minor safety earlier this year, so far its follow-through has been lacking—and since its transparency reports are not archived, that promise appears to be gone, too.

4.3 Transparency Audits

4.3.1 Providers vs NCMEC. Figure 5 shows the mean difference plots for the seven service providers that released CyberTipline report counts, first individually for each provider across seven panels and then for all providers in the eighth panel. For each pair of report counts disclosed by a service provider and NCMEC, the mean difference plots include a point with x the mean of the two counts, y the difference of the two counts as a percentage of the mean, and the color reflecting the year. If the service provider’s count is smaller than NCMEC’s, the percent difference is positive (and vice versa). The dashed line is the mean of the percentage differences. Each of the eight plots has its own dimensions for axes and the eighth plot’s x-axis is logarithmic. Tables 3–5 in Appendix A provide the raw data.

Finding 10. Aylo’s Pornhub is the only service provider to acknowledge and explain a mistake in their handling of CyberTipline reports.

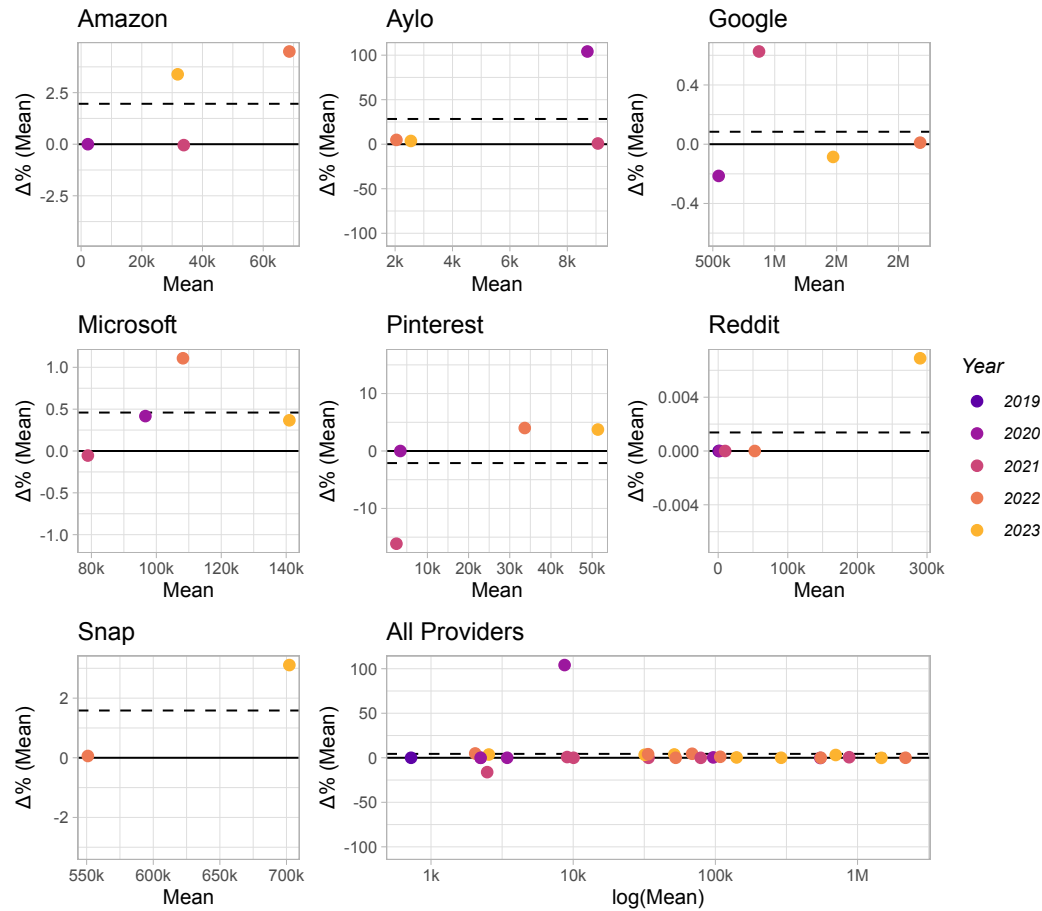


Fig. 5. Mean difference plots for seven service providers individually and all of them together. Each plot has its own scale, and the x-axis is logarithmic for the eighth plot.

When considering y-axis scales, the plot for one provider sticks out: Aylo's 104.1% difference for 2020. The firm only makes transparency disclosures for its most visible brand, Pornhub. In its transparency report for the year, which was published in April 2021, the brand not only acknowledges the discrepancy but also explains it as the result of submitting the same reports "multiple times in an abundance of caution" [107]. They add that NCMEC has been notified of the duplicate reports. Strikingly, this is the only instance we could find of a service provider acknowledging a problem with their disclosures or providing a credible explanation for the problem. While we consider the acknowledgment and explanation exemplary, the rest of the report makes for tedious reading. It emphasizes, again and again, Pornhub's principled stand against any form of abuse.

Presumably, the firm is still reacting to events from December 2020, when New York Times columnist Nick Kristof [73] made common cause with anti-porn crusaders [56] associated with a White supremacist church [49] and used a few harrowing cases to falsely paint Pornhub as the epitome of abusive sleaze [7, 45], a site "infested with rape videos."

No matter that similar cases regularly occur on social media as well. No matter that Pornhub had been working with NCMEC since 2019 and was proactively scanning user-contributed content already. No matter that Apple is not scanning proactively to this day. No matter that Kristof had published similarly sensationalist and ultimately false reports twice before [21, 121] and keeps casting himself in the role of the superhero rescuing damsels from monsters [53].

Finding 11. For the seven service providers that disclose at least a year of CyberTipline report counts, almost all of their counts are within 5% of NCMEC’s disclosures. They also exhibit a very small but statistically relevant bias towards underreporting.

Other than Aylo’s outlier and a considerably smaller one at -16.1% for Pinterest, all percent differences have a magnitude smaller than 5%. That is larger than the 1% threshold we set before data analysis (see Section 2.2). But it still is reasonably small. While that points towards a good faith effort at data collection and disclosure by those seven service providers, the mean difference plots make another pattern clearly discernible, namely that the majority of percentage differences is positive. That holds for individual values, with 16 positive, 6 zero, and 5 negative percentage differences. That also holds for the means of the percentage differences, which are indicated by a dashed line in each panel and positive for six out of seven providers as well as for all providers together. Even when disregarding the two outliers, the remaining 25 percent differences still have a positive mean of 1.2%. The sign test further validates that this bias is statistically significant with a p-value of 0.0133.

In theory, positive percentage differences are either the result of providers undercounting or NCMEC overcounting. In practice, NCMEC has little to gain by inflating report counts. After all, its role as clearinghouse for CyberTipline reports is legally mandated, and it receives federal funding for that function. By contrast, service providers have strong reputational and marketing incentives to make their platforms appear “safe,” lest they upset advertisers or (European) regulators [114]. It would appear that these pressures are reflected in the data. However, at 1.2% on average, the effect is small enough so it doesn’t distort the data in a meaningful way.

4.3.2 Providers Over Time. For the previous audit, which compared counts tabulated by independent organizations, a small amount of random error is not really surprising. For this, the second audit, that is emphatically not the case. In fact, we only considered the possibility after serendipitously using Meta’s csv file for a subsequent quarter while validating a code refactoring. Except it didn’t feel it at the time, as we spent the better part of a day checking and rechecking our code for bugs—until finally concluding that it had to be the input.

Finding 12. In addition to only including rounded quantities, Meta’s transparency data is marred by seven consecutive quarters with an average of 3.6% divergent entries over the previous quarter. Divergent entries vary widely in metrics and quarters, with no apparent temporal or logical locality.

For seven quarters, from Q3 2021 to Q1 2023 (inclusive), Meta’s csv file diverges in an average of 3.6% and range of 0.6%–7.3% of all entries from the previous quarter’s csv file. Divergent entries vary widely in policy area and actual metric (e.g., content actioned, appealed, or restored). Notably, 12, 3, 7, 7, 11, 17, and 4 entries about CSE differ from the previous quarter over that time range. Divergent entries also vary widely in the affected period, going back up to 7–11 quarters every quarter but Q4 2021 and Q1 2023. The latter two quarters also are the quarters with the fewest divergent entries of 1.2% and 0.6%, respectively. Overall, there is no discernible temporal or logical locality to these errors. Meta never acknowledged these discrepancies. Though they did stop just as the EU’s Digital Services Act was taking effect.

Worse, the actual extent of divergent entries across Meta’s transparency data is unknowable. Since Meta is the only service provider to round its transparency data, additional disparities may very well have been hidden by the rounding

process. By the same argument, we cannot exclude that Meta’s data continues to suffer from quality issues. They may just be small enough not to survive rounding. The presence of disparities also raises the question of why Meta decided to round transparency statistics in the first place.

Meta fared better in two previous audits of its “community standards enforcement report,” which the firm initiated itself and which did not find any significant short-comings. First, a panel of academic experts performed a qualitative review of Meta’s policies, enforcement process, and transparency reporting including metrics [6]. Second, Ernst & Young performed a quantitative assessment of Meta’s transparency data collection according to usual accounting standards [84]. According to E&Y’s written statement, the accountants reviewed Meta’s management assertion about the transparency data and found that “internal controls were suitably designed, implemented and operating with sufficient effectiveness.” Now, E&Y only reviewed Q4 2021, and that quarter had the second lowest rate of divergent quantities, with $27/2,192 = 1.2\%$ of data points. Still, their assessment clearly wasn’t sufficiently thorough.

While Meta has certainly invested significant resources into trust and safety, the firm also has a substantial history of faulty and misleading metrics and favoring profits over minor safety. Notably, it misreported advertising metrics to customers [8, 59–61, 136], platform usage data to a Harvard-based research consortium [130], and adverts in its transparency database [113, 117, 120]. Meta also deplatformed the academics who collected the data behind several of the news reports just cited [25, 26] and shut down its CrowdTangle transparency tool when it became inconvenient [70]. While employed by Meta, the author also discovered compelling evidence that the firm was artificially inflating basic advertising metrics [46]. As internal documents unsealed as part of a lawsuit by the State of New Mexico against Meta demonstrate [126], Meta running an advertising campaign (for TikTok nonetheless) that sexualized teenage girls for the benefit of middle-aged straight men was not an isolated incident [119].

Finding 13. TikTok’s transparency data on CSE is mostly unusable. Its csv files are marred by differences in label casing and floating point serialization. Claims made in its Q2 2023 transparency report about improved disclosure granularity are false when it comes to minor safety.

TikTok discloses absolute counts only for the number of actioned videos; all other metrics such as the breakdown into categories and subcategories of violative content are expressed as decimal fractions. However, before the firm’s introduction of a new schema in Q2 2023, the firm did not disclose all subcategory fractions, notably omitting the fraction of videos actioned by automation, and therefore made it impossible to calculate the number of videos actioned because they contain CSAM. The firm claimed that the new schema “added granularity.” With an increase from 26 to 29 subcategories, that is technically true. But in practice, TikTok also removed granularity, notably reducing the number of subcategories for minor safety from five to four, which are too coarse, again making it impossible to determine the number of videos actioned because of CSAM. Meanwhile, the firm’s csv files are marred by inconsistent capitalization for labels and inconsistent serialization of floating point numbers, e.g., the same quantity may appear as “0.9620000000000001” and “0.962.”

The lack of complete statistics when it comes to CSE suffices for rendering TikTok’s transparency disclosures useless for our purposes. However, even if it was disclosing the necessary metrics, the subpar state of their csv files would unnecessarily complicate analysis.

5 DISCUSSION

Knowing that much of the growth in CyberTipline reports is driven by an equivalent growth in social media users doesn’t make NCMC’s work any easier. The organization is already close to breaking point [48]. But at the current rate

of 3.93 million more reports per year, it will need to process an additional 7.5 reports for every minute in every year. At some point in the future, growth will necessarily plateau. After all, there are only so many humans who can maintain only so many social media accounts. Alas, assuming that children under 10 mostly stay off social media and the rest of humanity maintain two social media accounts on average (compared to six for this study's author), that still implies $2.74\times$ the current CyberTipline report volume. Hence reducing the volume of CyberTipline reports is imperative.

5.1 Legal Overreach

This study suggests two avenues for meaningfully reducing the CyberTipline report volume. The first is legislative reform based on the realization that, in the majority of cases, sexual imagery involving minors does not involve sexual predation. That makes the blanket prohibition against such imagery excessively punitive. It also impedes the self-determination of adolescents. While the United States is one of only two countries in the world that hasn't ratified the United Nations Convention on the Rights of the Child [83] and thus does not recognize freedom from excessive interference as a right for children, the blanket prohibition does interfere with other US laws.

For instance, in roughly half of US states, the age of consent is 16 or 17. By definition, that allows for consensual sexual relationships between an adolescent and an adult. However, if the adult takes (consensual) pictures of their interactions and shares those pictures with their partner, the adult not only violates the prohibition against CSAM but also meets the Sentencing Guidelines' conditions of sending violative images to a minor, thus facing a *minimum* penalty of 87–108 months in prison—for privately sharing pictures of a consensual and legal activity with the other participant. All it takes for police to get involved is for one of the minor's parents, who happens to disprove of the relationship, to look through their child's cell phone pictures. That threat is particularly acute for LGBT youth, who encounter substantial discrimination in the criminal justice system [118].

Unfortunately, NCMC and other advocacy groups are aiding this legal overreach by referring to sexual imagery created by and involving minors as “self-generated child sexual abuse materials” or shorter “self-generated CSAM.” It strongly suggests that these organizations are unable to conceive of adolescents as sexual beings just when they are particularly focused on just that aspect of their rapidly maturing bodies. That, in turn, gets in the way of supporting adolescents in navigating what activities are and are not acceptable, notably, when sexting. When all imagery is verboten, working out the practical (and moral) differences becomes much harder. The fact that non-consensual sharing has a substantial prevalence $> 10\%$ suggests that distinguishing between the two is, in fact, challenging for many adolescents.

None of the above is an argument for legalizing the public posting or arbitrary exchange of sexual imagery involving minors. But these scenarios serve as compelling arguments for *selectively* eliminating the attendant criminal jeopardy. It may also be worth considering whether the exchange of inappropriate memes or expressions of abhorrence are better treated as something akin to traffic violations. Since the mid-1990s, the US spent about \$2 billion on abstinence-only sex education, with nothing to show for that investment besides “increasing adolescent birthrates in conservative states” [40]. It would be pure folly to keep trying this same failed strategy for sexting.

5.2 With the Deluge

The second second avenue for reducing the deluge of reports is undoing the growth of normalized yearly rates from one to eight reports per 1,000 user accounts. Compared to the other statistics in this paper with their millions to thousands of millions, these are human-conceivable dimensions. While seemingly larger than previously reported rates for offline CSE, they nonetheless are inconsistent with the notion that the internet activated a previously latent human potential

for sexual predation. Still, the increase from one to eight reports per 1,000 user accounts over the last decade is very clearly noticeable.

Making sense of this component requires additional research—and more granular data including on the prevalence of sexts, memes, and predatory materials. However, several factors point towards social media being responsible. There is the finding that most of the growth is driven by social media user accounts. There is the “move fast and break things” mindset of US-based corporate social media and their relentless pursuit of hypergrowth. Not surprisingly, the resulting platforms make posting of user-generated content easier than any other platform before and are designed for the viral distribution thereof. Facebook contributing to the genocides in Myanmar and Ethiopia are extreme examples for what can go wrong as a result [43, 44, 58, 92]. Finally, there is commentary by observers of internet culture that attests to 2014 as the year when social media displaced the open web [13, 122].

While suggestive, more work is clearly needed to make sense of this phenomenon let alone undo it. In the meantime, more detailed reporting by the technology industry can make a significant difference so that NCMEC can continue serving as an effective global clearinghouse for reports about CSE.

5.3 Further Failures of Transparency

In addition to the audits, the process of selecting and evaluating service providers in preparation for this study surfaced concerns about the transparency practices of a few more firms.

When Discord first disclosed statistics about its CyberTipline reporting in its transparency report for July to December 2020 [22], it did so in a section titled “Reports to NCMEC” in a bar chart titled “H2 2020 NCMEC Reports.” Based on this labelling, we started tracking Discord’s statistics as CyberTipline reports. Subsequent transparency reports did not use the same section headers, but still labelled the dominant statistic in their breakdown of triggers as “media reports” (as opposed to grooming, a considerably smaller category). During analysis, Discord’s statistics stuck out: They were considerably smaller than NCMEC’s counts and, according to Fisher’s exact test, definitely not random outliers. When we revisited the transparency report, we noticed a boldface “6,948 *accounts* to NCMEC” (emphasis ours) in the paragraph below the chart. It is eventually followed by a disclaimer that the “figure represents distinct accounts reported and not the total number of reports submitted.”

We briefly considered listing Discord as one of the audited firms and then revealing the mistaken metric in the analysis, thus drawing attention to the real-world impact of such misleading disclosures. But by doing so, we would have recreated the conditions of Discord’s disclosure, i.e., hiding critical information in the body text. Needless to say, that would be highly inappropriate for a scholarly article. Though it might provide a good starting point for a classroom exercise.

Discord is not the first technology firm to replace CyberTipline report counts with reported account numbers. Pre-Musk Twitter did so as well. It shifts the focus from measuring (some approximation of) incidents to measuring (some approximation of) disruptive individuals. It also reduces the magnitude of disclosed quantities. For instance, Discord’s flagged accounts are between 34% and 83% of the number of reports NCMEC received from the firm. That may make this metric more attractive to comms and marketing. But it also calls into question the firms’ often emphasized “zero tolerance” when it comes to the sexual exploitation of minors. After all, if the number of accounts is smaller than the number of incidents, some of those accounts must have caused several incidents. Since zero tolerance implies that offending accounts are immediately deactivated on discovery of the first incident, the question becomes how did they cause additional incidents? Were they discovered after the fact because of increased account scrutiny including history? Or how do you account for this discrepancy?

Table 2. Summary of surveyed organizations and their transparency practices.

<i>Providers That Make...</i>	<i>Count</i>	<i>Grade</i>
Reasonably accurate disclosures	7	Pass
Not enough disclosures (yet)	1	
Problematic disclosures	3	Fail
No disclosures	5	
<i>Total</i>	16	

We have observed more dubious behavior as well. Reddit stands out amongst the surveyed providers because it is the only one that has been disclosing CyberTipline report counts since 2019. Furthermore, its disclosed counts are the same as NCMEC’s for four out of the last five years. In the past, Reddit has committed to providing meaningful notice about its content moderation decisions and also to abiding by the Santa Clara Principles for content moderation [109]. Those commitments helped make it the only technology platform to receive five stars in the Electronic Frontier Foundation’s Who Has Your Back transparency rankings [16]. In short, Reddit appears to be exemplary when it comes to transparency. Except, as we discovered during the research for this study, Reddit has been shadow moderating comments for years, displaying removed comments only to their authors [52]. That, of course, is the opposite of meaningful notice and also a direct violation of the Santa Clara Principles.

Of course, there is a simpler option available to technology firms: Skip transparency disclosures altogether. In fact, Apple goes even further, and after its attempt at client-side scanning for CSAM failed [41], the firm gave up on the issue, forgoing even cloud-based scanning. While pervasive encryption probably would limit what content can be scanned, Meta’s WhatsApp also faces that problem and nonetheless managed to file 1.4 million CyberTipline reports in 2023. Instead, Apple chooses not to scan for CSAM and not to make transparency disclosures, a degree of inaction only shared with Quora and possibly Omegle amongst surveyed organizations. Doing so is legal—as long as Apple files CyberTipline reports about incidents as it becomes aware of them, which it did 267 times in 2023. Only the Wikimedia Foundation filed fewer reports.

5.4 Incentives for Improving Transparency

Table 2 captures a final tally of organizations (other than NCMEC) across four transparency disclosure categories. First, all seven firms participating in the first audit make reasonably accurate disclosures. Second, X has only released a half year’s worth of transparency data and hence could not yet be audited. Third, Discord, Meta, and TikTok have made problematic transparency disclosures that render some or all of their data useless. Fourth, five more organizations skip transparency disclosures about CSE altogether. In other words, one half of surveyed providers pass their reviews. The other half of surveyed providers fail their reviews. That may be better than we expected, but it also does not boost confidence in the technology industry’s ability to understand let alone moderate their social impact.

One obvious remedy for providers failing their transparency reviews is a legal mandate that compels firms to track and disclose specific metrics. Alas, when taking the results of the 2024 US presidential election into account, that remedy also appears unrealistic for the foreseeable future. At the same time, this study illustrates the very benefits of such a mandate, one anchored in US law nonetheless. Before the April 2023 release of NCMEC’s transparency report to the Congressional Appropriations Committees [99], several statistics including long-term growth in CyberTipline reports, kinds of reported incidents, number and uniqueness of attachments, and relationship between perpetrators and victims

were not publicly available. However, thanks to an addition to the explanatory statement for the 2023 Appropriations Bill [20], NCMEC was compelled to produce the corresponding transparency data. It made a real difference for this study.

Even if that particular stick remains out of play, we strongly believe that upping their game in transparency reporting is in the best interest of both industry and NCMEC as well. We belabored the reasons for needing more granular data in Sections 5.1 and 5.2 already. So here, we add only two more points. For one, social media firms probably won't escape the Brussels effect [5] and already are subject to the Digital Services Act's transparency reporting requirements [28]. They might as well do the same reporting on a global scale. For another, a lack of data provides a fertile breeding ground for various stakeholders' biased and simplistic accounts when the reality, as this study illustrates for CSE, often is far more nuanced and complicated. Admittedly, better data doesn't automatically result in more nuanced story telling (whether by press, NGOs, or academics), far from it. But it at least allows for that to happen.

For trust and safety professionals trying to make such transparency disclosures, we have three actionable recommendations based on our experiences with curating transparency data. First, strongly prefer absolute metrics over relative ones. People working with the data can easily turn absolute quantities into normalized, fractional ones, whereas the reverse transformation usually is harder and loses precision. Second, disclose all metrics, including historical quantities, without rounding, in csv format with release notes. Keeping comms or marketing out of transparency reporting may be too much to ask for. But please limit their spin to the written report with pretty graphs. Third, close the loop, for example, by having freshly hired data scientists write reports based on the available data. Then incorporate their feedback.

6 CONCLUSION

In summary, the evidentiary record does *not* support the more sensationalist claims about child sexual exploitation. There is *no* "explosion" of CSAM. The internet is *not* "overrun with images of child sexual abuse." The pandemic did *not* trigger more sexual exploitation overall, even if there was a marked spike in March 2020.

Nonetheless, the evidentiary record does support the claim that US-based social media with their unprecedented global reach and propensity for virality, the near total legal prohibition against sexual imagery involving minors, and perfectly normal adolescent exploration of sexuality, boundaries, relationships, and so on have combined to create a perfect storm of CyberTipline reports. And that storm is now threatening to overwhelm the very organization serving as a critical global resource. Not only that, but the majority of reports is also useless. That should suffice as encouragement for legislative reform. The fact that laws intended to protect minors are increasingly criminalizing and thereby harming them should add urgency. That also makes language such as "self-generated CSAM" counter-productive.

As far as transparency is concerned, many technology firms did far better than we expected. As this study has amply demonstrated, transparency disclosures needn't be mere transparency theater, but can help produce real insight. The most surprising result from this study is the first audit's finding that firms generally disclose reasonably accurate CyberTipline report counts but nonetheless have a very small but statistically significant bias towards underreporting. In other words, the professionalism of the trust and safety personnel preparing these statistics prevails, but it cannot entirely overcome all too human biases. Making such discoveries requires access to meaningful data. As this study has shown, we do not have enough when it comes to online sexual exploitation of children.

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A THE DATA

Tables 3–5 provide the data released by electronic service providers and NCMEC about CyberTipline reports and attached pieces, aggregated by corporation and year. Parenthesized material after provider names repeats the disclosure frequencies from Table 1 as Q/H/Y for every 4/6/12 months, respectively, and then adds picture and/or video icons if the provider proactively scans for CSAM. Pinterest is marked “Q/H” because the firm releases quarterly aggregates every six months.

The $\Delta\%$ column shows the percentage difference between the two report counts in terms of the mean, just as in Figure 5. Some of the entries are, in fact, identical; the corresponding entries are marked as $\equiv$. Also, some of the percentages for Quora, Apple, and the Wikimedia Foundation in the middle column for NCMEC’s data are too small to fit into the allotted three-digit precision and hence are marked as ϵ .

Since years in the first column and total report counts in the last column are repeated for every service provider, they are presented in gray to make tables look less busy. The open space for providers that do not make transparency disclosures strengthens that effect. Alas, the motivation for giving every provider the same table real estate was to avoid biasing the presentation.

Providers are sorted in descending order of report counts for 2023.

Table 3. CSAM pieces and CyberTipline reports disclosed by technology firms and NCMEC: 2019 to 2023 (pt. 1)

	Disclosed by Service Provider					Disclosed by NCMEC		
Year	Pieces	per	Reports	Δ%	Reports	of (%)	Total	
META (Q 2024)								
2019	39,368,400	2.48			15,884,511	93.508	16,987,361	
2020	38,890,800	1.92			20,307,216	93.362	21,751,085	
2021	78,012,400	2.90			26,885,302	91.454	29,397,681	
2022	105,800,000	3.89			27,190,665	84.814	32,059,029	
2023	63,300,000	2.07			30,658,047	84.666	36,210,368	
GOOGLE (H 2024)								
2019					449,283	2.645	16,987,361	
2020	4,437,853	8.10	547,875	-0.21	546,704	2.513	21,751,085	
2021	6,696,497	7.69	870,319	+0.63	875,783	2.979	29,397,681	
2022	13,402,885	6.16	2,174,319	+0.01	2,174,548	6.783	32,059,029	
2023	7,955,169	5.40	1,472,221	-0.09	1,470,958	4.062	36,210,368	
X NÉE TWITTER (H 2024)								
2019					45,726	0.269	16,987,361	
2020					65,062	0.299	21,751,085	
2021					86,666	0.295	29,397,681	
2022					98,050	0.306	32,059,029	
2023					870,503	2.404	36,210,368	
SNAP (H 2024)								
2019					82,030	0.483	16,987,361	
2020					144,095	0.662	21,751,085	
2021					512,522	1.743	29,397,681	
2022	1,273,838	2.31	550,755	+0.06	551,086	1.719	32,059,029	
2023	1,594,805	2.31	691,225	+3.11	713,055	1.969	36,210,368	
TIKTOK (Q 2024)								
2019					596	0.004	16,987,361	
2020					22,692	0.104	21,751,085	
2021					154,618	0.526	29,397,681	
2022					288,125	0.899	32,059,029	
2023					590,376	1.630	36,210,368	
DISCORD (Q 2024)								
2019					19,480	0.115	16,987,361	
2020					15,324	0.071	21,751,085	
2021					29,606	0.101	29,397,681	
2022					169,800	0.530	32,059,029	
2023					339,412	0.937	36,210,368	
REDDIT (H 2024)								
2019			724	≡	724	0.004	16,987,361	
2020			2,233	≡	2,233	0.010	21,751,085	
2021	9,258	0.92	10,059	≡	10,059	0.034	29,397,681	
2022	80,888	1.54	52,592	≡	52,592	0.164	32,059,029	
2023			290,121	+0.01	290,141	0.801	36,210,368	

Table 4. CSAM pieces and CyberTipline reports disclosed by technology firms and NCMEC: 2019 to 2023 (pt. 2)

	Disclosed by Service Provider					Disclosed by NCMEC		
Year	Pieces	per	Reports	Δ%	Reports	of (%)	Total	
OMEGLE								
2019					3,470	0.020	16,987,361	
2020					20,265	0.093	21,751,085	
2021					46,924	0.160	29,397,681	
2022					608,601	1.898	32,059,029	
2023					188,102	0.520	36,210,368	
MICROSOFT (H )								
2019					123,927	0.730	16,987,361	
2020	1,256,652	13.03	96,435	+0.42	96,836	0.445	21,751,085	
2021	564,383	7.12	78,926	-0.05	78,883	0.268	29,397,681	
2022	452,384	4.22	107,599	+1.11	108,798	0.339	32,059,029	
2023	402,630	2.86	140,720	+0.37	141,236	0.390	36,210,368	
PINTEREST (Q/H )								
2019					7,360	0.043	16,987,361	
2020			3,432	≡	3,432	0.016	21,751,085	
2021	1,608	0.60	2,684	-16.15	2,283	0.008	29,397,681	
2022	37,136	1.13	32,964	+4.00	34,310	0.107	32,059,029	
2023	57,774	1.15	50,437	+3.73	52,356	0.145	36,210,368	
AMAZON (Y )								
2019					549	0.003	16,987,361	
2020			2,235	≡	2,235	0.010	21,751,085	
2021	27,244	0.81	33,848	-0.04	33,833	0.116	29,397,681	
2022	52,656	0.79	67,073	+4.49	70,157	0.220	32,059,029	
2023	24,756	0.79	31,281	+3.39	32,359	0.090	36,210,368	
AUTOMATTIC ()								
2019					10,443	0.062	16,987,361	
2020					9,130	0.042	21,751,085	
2021					4,821	0.016	29,397,681	
2022					5,035	0.016	32,059,029	
2023					19,591	0.054	36,210,368	
QUORA								
2019					1	€	16,987,361	
2020					2	€	21,751,085	
2021					25	€	29,397,681	
2022					2,242	0.007	32,059,029	
2023					6,135	0.017	36,210,368	
AYLO NÉE MINDGEEK (H )								
2019							16,987,361	
2020			4,171	+104.12	13,229	0.061	21,751,085	
2021	20,401	2.26	9,029	+0.82	9,103	0.031	29,397,681	
2022	9,588	4.80	1,996	+4.84	2,095	0.007	32,059,029	
2023	7,313	2.92	2,503	+3.65	2,596	0.007	36,210,368	

Table 5. CSAM pieces and CyberTipline reports disclosed by technology firms and NCMEC: 2019 to 2023 (pt. 3)

	Disclosed by Service Provider			Δ%	Disclosed by NCMEC		
Year	Pieces	per	Reports		Reports	of (%)	Total
APPLE							
2019					205	€	16,987,361
2020					265	€	21,751,085
2021					160	€	29,397,681
2022					234	€	32,059,029
2023					267	€	36,210,368
WIKIMEDIA (2)							
2019					13	€	16,987,361
2020					11	€	21,751,085
2021					8	€	29,397,681
2022					29	€	32,059,029
2023					34	€	36,210,368